

Measurement Inventory for Study Planning (formerly called Imaging Inventory for Study Planning)

SAS Dataset Name: MeasInventory

General description:

The purpose of this dataset is to facilitate planning of analyses using the OAI images and/or central readings from OAI images by compiling into one dataset information for every participant at each visit from baseline through 96 months: 1) the images that are in the public inventory (e.g. radiographs by type and MRI by acquisition); 2) the central readings and measurement from study images that are in the public datasets; and 3) the biospecimen assays that are in the public datasets. This will make study planning and participant knee/image selection easier than if multiple different databases have to be searched to determine if the images and reading data needed for a study are available.

IMPORTANT NOTE: The Measurement Inventory dataset does not contain information about data included as part of the OA Biomarkers Consortium FNIH Project.

The dataset consists primarily of indicator variables for each image type, by visit, and for the central reading data available for those images, also by visit. For example, the dataset contains an indicator variable for whether a participant has a Coronal IW TSE MRI scan in the right knee at 24 months and an indicator variable for whether a participant has data from central image assessment Project 09 (quantitative cartilage measurements using the sagittal DESS scan) at 12 months. Detailed descriptions of the data in each central image assessment project can be found in the descriptive documentation that accompanies the corresponding compressed data file.

The dataset also contains the ID variable, which is needed to link this dataset to other OAI datasets, and variables from other datasets for quick reference to information that will be useful in identifying participants and knees to be selected for an analysis. These latter include baseline Kellgren Lawrence grades from central readings, where available, and baseline knee pain status, whether the participant attended a clinic visit, cohort assignment and other data useful for selecting a study sample.

Structure. The dataset consists of one row/observation per participant.

IMPORTANT NOTE: Additional data about images and central reading data will be included in this dataset, and analysts are encouraged to ensure they are using the most up-to-date version of the dataset. Analyses ran using an older version of this dataset should be re-run using the most recent version of this dataset.

Description of main variables:Central Image Assessment Projects.

The dataset includes central image assessment project indicator variables for each image assessment project that has been released. These variables are project, visit, and side specific, and indicate if a participant is included in the project or not (possible values are 'Y' for Yes, 'N' for No).

Naming convention. In Cxx_yy_z, where xx indicates the project number, yy indicates the visit, and z indicates the side. For example:

In C03_BLL: indicates whether a participant's left knee was included in the central image assessment Project 03 at the baseline visit;

In C60_12mR: indicates whether a participant's right knee was included in the central image assessment Project 60 at the 12-month visit.

Please see the “*Overview and Description of Central Image Assessments*” document for detailed information about image assessment data and projects.

Radiograph Images.

Variables exist that indicate if certain types of x-rays have been released for specific visits for participants (possible values are 'Y' for Yes, 'N' for No).

Naming convention. xx_yy, where xx indicates the type of x-ray (full-limb x-ray, pelvis x-ray, or knee x-ray), and yy indicates the visit. For example:

Xray_BL: indicates whether a bilateral fixed-flexion knee x-ray is available from the baseline visit;

FullLimb_12m: indicates whether a full-limb x-ray is available from the 12-month visit. The fixed flexion, full limb and pelvis x-rays are bilateral, so side is not indicated.

MR Images.

There are variables that indicate whether certain MR Images have been released from specific visits for participants (possible values are 'Y' for Yes, 'N' for No).

Naming convention. The variable naming convention is MRI_xxyzz, where xx indicates the type of MRI sequence, y indicates the side, and zz indicates the visit. For example: MRI_AXMPRL12m: indicates whether the AX MPR sequence is available for the left knee from the 12-month visit.

IMPORTANT: Analysts are encouraged to view the variable label which contains additional information on which MRI sequence is associated with the variable name.

Variables also exist that indicate if the core set of MRIs is available for a knee from the specified visit (possible values are 'Y' for Yes, 'N' for No). The core set of MRIs is defined as the: COR IW TSE, SAG IW TSE FS, SAG 3D DESS WE, AXIAL MPR, and COR MPR acquisitions.

Naming convention. MRI_Corexy where x indicates side and y indicates the visit. For example, MRI_CoreL12m indicates whether the core set of MRIs was acquired on the left knee at the 12-month visit.

Other variables. As of release version 9, the RAVISIT variable was removed from the dataset, and replaced with RADX_Visit. RADX_Visit reports the first visit at which a participant reports having been diagnosed with rheumatoid arthritis (RA) and reports RA medication use. RADX_Visit is based on RADX (follow-up visit RA self-report diagnosis and RA medication use) when RADX=2, 3, or 4. The RADX variable can be found in the Medical History and AllClinical datasets. **IMPORTANT NOTE ABOUT RA:** Some participants have developed RA during the course of the study. Users should exclude participants with RA from analyses when appropriate using the RADX_Visit variable or the RADX variable.

The variables IndexKneeR and IndexKneeL indicate whether right and left knees, respectively, met the eligibility criteria to be included in OAI Core Image Assessment Sample from which Project 09 knees were sampled. A 'Y' indicates that the knee meets the eligibility requirements. These requirements were that the participant:

- was assigned to the Progression cohort
- had frequent knee symptoms at baseline (P01RK SX=2 or P01LK SX=2)
- had a baseline Kellgren Lawrence grade of 2 or 3 (based on central assessment whenever available)
- had a complete set of core MRI acquisitions at the baseline and 24-month visits in the same knee (MRI_CoreRBL/MRI_CoreLBL and MRI_CoreR12m/MRI_CoreL12m) and a fixed-flexion knee x-ray was available at the screening and 24-month visits (Xray_BL and Xray_24m)

The baseline visit central **Kellgren and Lawrence grade** variables for the right and left knees (V00XRKLR,V00XRKLL) included in the Measurement Inventory are taken from the baseline kXR SQ reading (BU) dataset (kXR_SQ_BU00). Data for these variables come from Project 37 if the knee was read for Project 37, or from Project 15. Detailed descriptions of the data in each central image assessment project can be found in the descriptive documentation that accompanies the corresponding data file.

IMPORTANT NOTE: Since the last versions of this dataset, values of V00XRKLR and V00XRKLL from Project 37 are used if they exist, rather than values from Project 15. The only situations where this has caused a change are in a few knees where the value has changed from 0 to 1, or vice versa.

The baseline **Kellgren and Lawrence grade equivalent** variables for the right and left knees (P01OAGRDR, P01OAGRDL) were calculated from the clinical center reading of

osteophytes and joint space narrowing. They can also be found in the Biomarkers00 or AllClinical00 datasets. More information about these grades can be found in [BaselineScreeningKneeXray OARSI KL.pdf](#). Note that in the Measurement Inventory dataset, the baseline calculated Kellgren and Lawrence grades are presented only in cases where the baseline central reading Kellgren and Lawrence grades (V00XRKLR, V00XRKLL) are missing.

The baseline symptom status variables for the right and left knees (P01RK SX, P01LK SX) can also be found in the JointSx00 or AllClinical00 datasets.

Description of variables from Outcomes dataset:

Knee and hip replacement status during follow-up. ExxRPCF is the key variable for identifying knee and/or hip replacements that occurred after a participant enrolled in the study. For participants who have reported a knee or hip replacement during their follow-up interviews, this variable gives information on where participants are in the adjudication process and has four possible values:

- ‘0: Replacement adjudicated, failed to confirm:’ this value is assigned when a reported replacement was determined through the adjudication process to not be a true replacement (e.g., surgery was for a different procedure). Participants for whom it was determined through the adjudication process that a reported replacement did not occur are generally not included in the Outcomes dataset unless they are in that dataset for another reason (e.g., replacement on contralateral side or other joint).
- ‘1: Self-reported replacement:’ this value is assigned when a knee or hip replacement is reported by the participant, the Coordinating Center has been notified that medical records cannot be obtained, and there are no x-rays for the visit(s) following the reported replacement from which to confirm the replacement.
- ‘2: Replacement pending adjudication:’ this value is assigned if a replacement has been reported and the adjudication process has not yet been completed, generally because the Coordinating Center is waiting to receive the medical records. This value may change in subsequent releases of the Outcomes dataset as medical record documentation is received and/or x-rays are acquired for these participants.
- ‘3: Replacement adjudicated, confirmed:’ this value is assigned if a replacement has been confirmed through the adjudication process from medical records and/or post-operative x-rays.
- ‘.A:’ this SAS special missing value code denotes that no replacement has been reported for that joint and side.

To select all participants with replacements that occurred after enrollment and confirmed through the adjudication process use the ExxRPCF= 3 for both sides. For instance, to select all participants with confirmed **knee** replacements use ERKRPCF= 3 and ELKRPCF=3 (right and left, respectively); to select all participants with **hip** replacements

use **ERHRPCF**=3 (right hip) and **ELHRPCF**=3 (left hip). To select participants with a confirmed replacement or those who have a self-reported replacement but whose replacement has not been confirmed through adjudication, use **ExxRPCF**=1, 2 or 3.

Baseline knee or hip replacements. If a replacement was seen on the *baseline* OAI x-ray, this is noted in the variable **ExxBLRP**. To select all participants with baseline replacements use **ExxBLRP**=1 for both sides for the joint of interest.

Knee or hip replacement seen on follow-up OAI x-ray. If a replacement has been seen on a *follow-up* OAI x-ray, this is noted in the variable **ExxRPSN** regardless of whether a replacement has been reported by the participant.

- ‘0: No replacement is seen on any available follow-up x-ray:’ this value is assigned when a knee or hip replacement has not been seen on a follow-up x-ray. In cases where the value of **ExxRPSN**=‘0’ and the confirmation/adjudication status variable **ExxRPCF** confirms a knee or hip replacement, it is likely that the last OAI x-rays acquired and reviewed for evidence of the joint replacement were acquired before the replacement occurred. The value for **ExxRPSN** may change in the future if additional x-rays are acquired for the participant.
- ‘1: Yes, replacement is seen on one or more available follow-up x-rays:’ this value is assigned if a replacement has been seen on a follow-up x-ray. When the confirmation/adjudication status variable **ExxRPCF** indicates the adjudication is still pending, in some cases a replacement may have already been seen on an OAI follow-up x-ray (**ExxRPSN**=‘1’). For most purposes these can be treated as confirmed joint replacements though details such as the date and reason for the replacement will not yet be available.
- ‘2: There are no follow-up x-rays for this knee (or hip):’ this value is assigned if there are no follow-up x-rays available for the joint.

To select all participants with follow-up replacements seen on x-ray use **ExxRPSN**=1 for both sides for the joint of interest.

In general, when the only information about a knee or hip replacement is self-reported and not adjudicated (**ExxRPCF**=1 or 2 and **ExxRPSN**=0 or 2), these joints should be treated as having had a replacement in analyses. Similarly, if analyses require participants who have not had a replacement, self-reported replacements should be removed.

Visit follow-up knee or hip replacement self-reported at. As of release version 3, the visit at which the participant self-reported the knee or hip replacement at is noted in the variable **ExxVSRP**.

Data Structure/Contents:

This dataset contains one record per participant. The variable uniquely identifying a record is ID, and the datasets are sorted by ID, which can be used to merge/join to data in other datasets.

A full list and description of all the variables contained in this dataset can be found in the contents.pdf.

Condition of data:

- **Known data errors:** None at this time.
- **Dataset strengths/weaknesses:** None at this time.
- **Data changes:**
 - As of version 3, one participant (9332345) had the left knee replacement visit reported at variable, V99ELKVSRP, change from 48-month to 60-month because although the replacement occurred after the 48-month visit, it was inadvertently coded as being reported at the 48-month visit.
 - As of version 7, the name of the dataset was changed from Imaging Inventory for Study Planning to Measurement Inventory for Study Planning.
 - As of version 8, some participants baseline Kellgren and Lawrence grades have altered from 1 to 0 or vice-versa (see note on Page 3 of this document).
 - As of version 9, the RAVISIT variable was removed from the dataset, and replaced with RADX_Visit (see note on Page 3 of this document).
 - As of version 11, the variables INC62_MCBONE (In McAlindon Bone Ancillary Study Project 62) and INBIO01 (In Biospecimen Assays Project 01) were added to the Measurement Inventory dataset.

General strategies for use:

When using with other datasets, merge/join by ID.

Analysts are encouraged to always output and view SAS variable labels in their entirety to ensure important information about the variables is not lost. The maximum SAS label length is 160 characters.